

Economic Intelligence & Financial Markets Platform

Official Architecture & Data Engineering Documentation

1. Project Overview

The Economic Intelligence Platform is an end-to-end enterprise Data Engineering solution. Its objective is to extract, process, and store macroeconomic indicators from the Central Bank of Brazil (SGS) and historical cryptocurrency data (Yahoo Finance), consolidating everything into a Data Warehouse (PostgreSQL) for consumption via an Interactive Dashboard (Streamlit).

2. Architecture & Infrastructure

The infrastructure is 100% containerized using Docker and Docker Compose, ensuring isolation and reproducibility. The orchestrated services include:

- economics_postgres_gold: Relational Database (PostgreSQL 15).
- economics_pipeline: Python extraction orchestrator (Pandas, Requests, yFinance).
- economics_dashboard: Analytical web application (Streamlit + Plotly).

Quality control is handled via CI/CD on GitHub Actions, which spins up the database and runs unit tests on every new commit to the main branch.

3. Ingestion Models (Extract & Load)

The pipelines operate with error handling, dynamic typing, and resilient connections:

1. Cryptocurrency (BTC-USD): Daily ingestion of closing price and trading volume.
2. Macroeconomics: Collection of IPCA (Monthly), Unemployment Rate (Series 24369), and the USD/BRL exchange rate.

Data is sanitized (drop_duplicates, dropna) before being loaded via SQLAlchemy.

4. Data Modeling (Gold Layer)

The Economics Gold database contains the following optimized tables:

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- gold_bitcoin_metrics: date (PK), close_price, volume.
- gold_economic_indicators: date (PK), indicator_name (PK), value.

The tables use direct relational schemas to facilitate performant analytical queries.